

THE UNIVERSITY
of EDINBURGH

Optimizing Fraud Detection in the UK Banking System Report

Operational Research MSc
Methodology, Modelling and Consulting Skills

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Executive Summary

In the banking sector, the prevention and detection of fraudulent activity are vital. Fraudulent transactions pose a significant threat to the financial health and reputation of financial institutions. Therefore, developing robust and efficient strategies to detect and prevent fraud is of utmost importance.

In an effort to tackle this issue, we have developed and compared three fraud detection models: the Baseline Model (Model 1), the Customer Probability Adjustment Model (Model 2), and the Calibrated Fraud Probability Model (Model 3). Each model represents an escalating level of complexity and refinement in its approach to fraud detection.

Model 1 serves as a baseline, using a straightforward optimization approach without adjusting probabilities. Model 2 advances on Model 1 by incorporating customer probability adjustment, leveraging past data to prioritize transactions based on customer behaviour patterns. Model 3, the most sophisticated, adds the fraud probability calibration to Model 2, further refining the model's predictive accuracy over time. The models performance is compared using the investigation success rate, which is the share of fraudulent transactions that have been investigated

Key Findings

Model 3 significantly outperforms other models and achieves a success rate of 75.5% in fraud cases detection (in contrast to 32.6% and 21.2% of Models 1 and 2, respectively) and a success rate of 77.3% in terms of money saved (in contrast to 62.9% and 48.6% of Models 1 and 2, respectively) while investigating only 19 % of all considered transactions, when tested on the first batch data (also called 'training set' in this report). Furthermore, the success rates in terms of cases are between 71 and 78 % for all five banks, demonstrating the performance stability of model 3.

Model 3 has also proven to be highly effective in utilizing in-house investigators and remarkably reducing the need for external investigators to an average of 2 to 3 external investigators per bank per month. Another strength of Model 3 is its high success rate in terms of discovered fraud cases among low-amount transactions (76.6 % for the first batch of data), while other models have not investigated the small transactions at all.

Despite the drop in success rate to roughly 40 % when the model was run in a real-life scenario without data about false negatives, the performance remained stable across banks, a low number of external investigators were employed and the model retained a relatively high success rate for low-amount transactions while investigating roughly 6 % of all considered transactions from the first batch data and 18 % from the second batch (also called 'test set' in this report).

The key findings can be summarized in the following list:

1. Fine-tuning customer probabilities and calibrating fraud probabilities are critical for attaining peak performance. Accurately assessing expected losses and integrating historical data into the model are pivotal factors that contribute to enhancing overall effectiveness.

2. The resulting solution is stable across different banks, strong in fraud detection among low-amount transactions, and manages to efficiently utilize in-house investigators with little need for hiring external investigators.

3. The resulting model remains stable and performs in the same manner in the limited information case, when only information about investigated cases is available.

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1 Introduction

The detection and prevention of fraud are crucial due to the substantial risks it poses to individuals, businesses, and the stability of financial systems. Within the UK, fraud has emerged as the predominant crime type, presenting a pressing concern. Police Scotland has witnessed a concerning surge in reported fraud cases, surpassing the five-year average by 46.8%.[1] This upward trend exemplifies the magnitude of the issue and necessitates immediate attention and effective countermeasures. Given the criticality and urgency surrounding fraud in the banking system, the development of a robust fraud risk management model is paramount.

The presented optimization model is based on a small-scale simulation of the UK banking system. This simulation enables us to emulate the complexities of fraud detection and investigation within a more manageable context. In this scaled-down system, our simulation includes several banks, each equipped with its own in-house team of investigators. However, due to their limited capacity, these banks may occasionally need to enlist the help of external investigators for certain cases.

In this report, we will outline the approach undertaken to develop the optimization model, considering factors such as customer probabilities, transactional and description probabilities, and the expertise of in-house investigators. Furthermore, we will present the findings of the fraud detection model, highlighting its effectiveness in minimizing fraud costs and maximizing the detection of genuine fraudulent transactions.

1.1 Main terms

- Fraud probability is an estimation of risk of a particular transaction to be fraudulent. It is calculated for each transaction as the product of customer, transactional and description probabilities.
- Expected loss is calculated for each transaction as the product of corresponding fraud probability and amount.
- Cost of investigation per transaction equals to the cost of hiring external investigator, if any is needed. In case the transaction is not investigated or investigated by internal team, the cost is zero.
- Investigation success rate is the share of fraudulent transactions that have been investigated.

2 Objectives

1. Formulate an optimization problem aimed at minimizing the combined expected losses resulting from fraudulent activities and the total cost of investigations, all within the confines of defined constraints.
2. Devise a learning strategy that maximizes the use of historical data to enhance the ability to distinguish between fraudulent and legitimate transactions and to evaluate expected losses.
3. Assess the performance of solution on the second batch data sample and in case only information about investigated cases is available.

3 Solution Approach

Our approach consists of two main components: an optimization model that makes decisions about which transactions to investigate, and a learning model that leverages the available historical data to update the predicted fraud probabilities used by the optimization model.

3.1 Approach overview

The flowchart in Figure 1 illustrates the daily investigation procedure. The learning model consists of three sub-components: update of fraud probability threshold, customer probability adjustment and fraud probability calibration. These are described in more detail in section 4.4. In order to assess the performance of the sub-components, three fraud detection models were employed:

- Model 1: only the update on fraud probability threshold is implemented.
- Model 2: customer probability adjustment is added on top of Model 1.

[1] Police Scotland Quarter 4 YTD Performance Report: April 2022 to March 2023

- Model 3: all three sub-components are included.

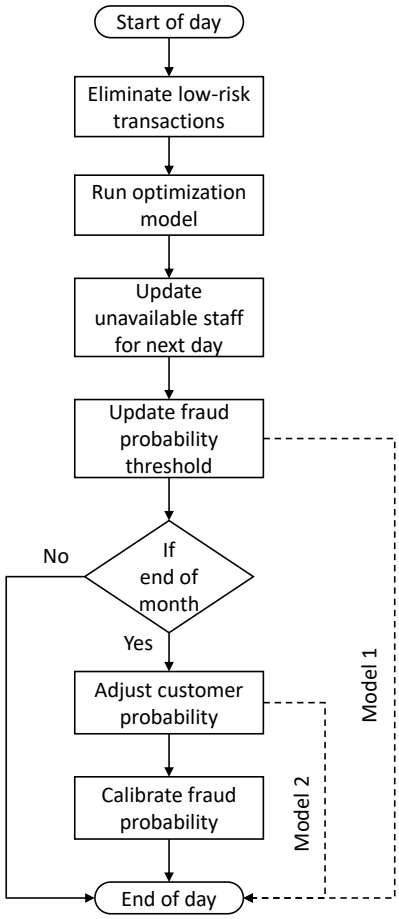


Figure 1: Flowchart of the daily process.

Step 1: Eliminate low-risk transactions

It is obvious that spending resources on low-risk transactions does not yield optimal results. Hence, transactions with fraud probability below the threshold are eliminated. Only the remaining transactions are considered by the optimization model.

Step 2: Run optimization model

The optimization model decides which transactions to investigate, and allocates them to internal investigators of each bank or external investigators that banks should employ. The goal is to maximize the expected gain from investigations, which is the difference between the expected amount of money recovered from fraudulent transactions and the cost of investigation.

Step 3: Update unavailable staff for next day

Since the investigation of priority 4 transactions takes two days, if an in-house investigator is assigned such an investigation, they will not be available the next day. Therefore, after the run of the optimization model, the number of unavailable staff in each bank is updated for the next day.

Step 4: Update fraud probability threshold

The threshold is dynamically updated at the end of each day based on historical data up to that day. It is calculated as the minimum of the 75th percentile in the probability distribution of non-fraudulent transactions and the 25th percentile in the probability distribution of fraudulent transactions.

Step 5: Adjust customer probability

We adjust the customer probability based on historical data up to that day. The new value corresponds to the probability of being scammed again given that a person has already been scammed. For customers who have never been scammed, the customer probability stays the same.

Step 6: Calibrate fraud probability

Using regression techniques, we calibrate the fraud probability with the goal of aligning it with the true fraud probability obtained from historical data on fraud cases up to that day. The fraud probability is calibrated separately for different transaction amount categories.

3.2 Assumptions

To build our optimization model for fraudulent transaction investigation, we've made several key assumptions that shape our approach and the design of the model.

- **Nature of fraudulent activity**

  Fraudulent activity primarily involves the unauthorized outflow of funds rather than inflow. Under this assumption, incomes and cash withdrawals are excluded from investigations.

- **Investigation scope**

  The bank only spends its resources investigating transactions in which it is involved. This assumption reflects the practical reality that the bank's control is limited to its own transactions. In cases where a transaction is between two UK banks, the investigation resources can be shared.

- **Investigation assignment**

  It's impossible to split an investigation between an external investigator and another investigator (either in-house or external). This assumption is based on the cost structure for external investigators, who are typically paid per case. If an external investigator is hired for a case, the cost remains the same regardless of whether they handle the entire investigation or only a portion of it.

- **External investigator limitation**

  Each bank cannot hire more external investigators than the number of in-house investigators it has. This limitation of reliance on external resources is requested by our client.

3.3 Optimization Model

On a daily basis, the optimization model decides which transactions to investigate based on their associated probabilities and potential gains. We formulate this as a Linear Programming (LP) problem, with the objective of maximizing the total expected gain from the investigations. The expected gain for each transaction is defined as the difference between the expected amount of money recovered from fraudulent transactions and the cost of investigation.

Model formulation

Index sets

I ... set of transactions on a given day (decided by probability threshold)
B ... set of banks

Input data

k_i ... priority of transaction i
p_i ... overall probability that transaction i is fraudulent (from learning model)
a_i ... amount of transaction i
r_{ib} ... $\begin{cases} 1 & \text{if bank } b \text{ is involved in transaction } i \\ 0 & \text{otherwise} \end{cases}$
f_b ... full investigation capacity of bank b
u_b ... number of internal investigators from bank b which are unavailable (from previous day)
m_b ... minimal number of in-house investigators in bank b that should be available on next day
c_{k_i} ... cost of hiring an external investigator for a transaction i with priority k_i

t_{k_i} ... investigation time (proportion of a day) for a transaction i with priority k_i

Decision variables

$$x_i = \begin{cases} 1 & \text{if transaction } i \text{ should be investigated} \\ 0 & \text{otherwise} \end{cases}$$

$$y_{ib} = \begin{cases} 1 & \text{if transaction } i \text{ requires an external investigator from bank } b \\ 0 & \text{otherwise} \end{cases}$$

z_{ib} ... investigation time of bank b spent on investigating transaction i (proportion of a day)

Objective function

We aim to maximize the expected net gain from investigations, which is given by the difference between expected amount of money recovered and the total cost of external investigations. In mathematical notations:

$$\max \sum_{i \in I} p_i a_i x_i - \sum_{i \in I} \sum_{b \in B} c_{k_i} y_{ib}$$

Constraints

1. External investigators can be used only if the transaction is investigated.

$$y_{ib} \leq x_i \qquad \forall i \in I, b \in B$$

2. Only banks involved in a transaction can hire external investigators to investigate it.

$$y_{ib} \leq r_{ib} \qquad \forall i \in I, b \in B$$

3. Only banks involved in a transaction can spend time investigating it

$$z_{ib} \leq t_{k_i} r_{ib}, \qquad \forall i \in I, b \in B$$

4. Amount of time spent on investigation for each transaction matches the priority requirement. Also, time is spent on investigation ($z_{ib} > 0$) only if transaction is investigated ($x_i = 1$).

$$\sum_{b \in B} z_{ib} = t_{k_i} x_i$$

5. Limit on in-house investigation time. The amount of time spent by internal investigators from each bank should not exceed the full capacity minus the number of available staff.

$$\sum_{i \in I} (z_{ib} - t_{k_i} y_{ib}) \leq f_b - u_b \qquad \forall b \in B$$

6. Limit on the number of external investigators.

$$\sum_{i \in I} y_{ib} \leq f_b \qquad \forall b \in B$$

7. In the case that external investigators are used, responsibilities cannot be split. (Only one y_{ib} can be 1 for all b. Also, if $y_{ib} = 1$, then $z_{ib} = t_{k_i}$.)

$$\sum_{b \in B} y_{ib} \leq 1 \qquad \forall i \in I$$

$$t_{k_i} y_{ib} \leq z_{ib} \qquad \forall i \in I, b \in B,$$

8. Control on priority 4 investigations: only f_b - m_b internal investigators can engage in priority 4 transactions. This ensures we have in-house investigators for the next day. For now, we choose $m_b = \frac{1}{2} f_b$. (Results show that this constraint do not matter due to the low amount of suspicious priority 4 transactions)

$$\sum_{i \in I | k_i = 4} (z_{ib} - y_{ib}) \leq f_b - m_b \qquad \forall b \in B$$

9. Non-negativity.

$$z_{ib} \geq 0, \forall i \in I, b \in B$$

3.4 Learning Model

Update of fraud probability threshold.

The fraud probability threshold is used to select transactions for further investigation. Only transactions with fraud probability higher or equal to threshold would be considered. A well-chosen threshold plays a significant role in our solution. With a threshold too low, there is a risk of over-investigation and making incorrect decisions about what transactions to investigate. With a threshold too high, there is a risk of omitting a majority of fraudulent transactions.

The threshold is dynamically updated at the end of each day based on historical data up to that day (excluding the current day's priority 4 investigated transactions, since their investigation is not finished until the next day). It is calculated as the minimum between the 75th percentile in the probability distribution of non-fraudulent transactions and the 25th percentile in the probability distribution of fraudulent transactions. The initial value for day 1 was set to 0.05 in order to accommodate more transactions.

Customer probability adjustment.

It has been shown that people who have fallen victim to a scam are more likely to be scammed again in the future [2]. Therefore, it is appropriate to adjust customer probability of customers who have been victims of scams. The customer probability provided in the model is representative of the likelihood of being scammed and does not inherently account for historical information. To address this limitation, a conditional update is implemented. If a customer has experienced being scammed at least once, their customer probability is adjusted to reflect this historical context. In other words, the customer probability becomes conditional, indicating the probability of being scammed given that the customer has already experienced a scam in the past. This adjustment allows the model to factor in the customer's history, making the probability more nuanced and tailored to individual experiences with fraudulent activities.

The calculation process is as follows: customers are grouped based on their original customer probability. Within each group, the number of customers who have experienced at least one scam is determined. Subsequently, for each group, the proportion of customers who have been scammed more than once is calculated relative to the total number of customers who have experienced scams. This calculated proportion serves as the basis for updating the customer probability, given both the original customer probability and the occurrence of repetitive scam cases. The result is an adjusted customer probability that incorporates information about repeated instances of scam within each probability group. Following the calculation process outlined, the customer probability is selectively updated for those customers who have encountered fraud cases. Specifically, the update is applied based on the resulting probability derived from the corresponding group to which each customer belongs. This targeted updating approach ensures that the adjustments only impact customers with a history of fraud, aligning the updated probabilities with the observed patterns within their respective probability groups. The customer probability adjustment is conducted on a weekly basis.

Fraud probability calibration.

The main aim of probability calibration is to overcome the shortcomings of machine learning models used to calculate the customer, transaction and description probabilities. Due to the nature of target values, the models have been trained on imbalanced datasets. Such models often tend to overestimate probabilities near zero values and underestimate near one, or vice versa. The consequence of this is that while the scores produced preserve order (the higher the number, the more likely the positive class), they cannot be interpreted as frequentist probabilities. As a result, the evaluation of expected losses would be biased as well.

The information from October, 2023 is decided to be utilized for analytical purposes. The reliability diagram serves as a key tool in this analysis. This diagram is a line plot that compares the relative frequency of observed events (depicted on the vertical axis) with the predicted probability frequency (depicted on the horizontal axis).

Given that the primary objective of calibration is to ensure that the fraud probabilities used in calculating expected losses align with the actual observed probabilities, a critical aspect of the analysis involves assessing the accuracy of predicted probabilities in relation to the transaction size. Transactions are categorized into four bins based on their amount: (1) less than or equal to £45, (2) more than £45 but less than or equal to £75, (3) more than £75 but less than or equal to £120, and (4) more than £120. These bins are strategically selected to roughly correspond to the 25th, 50th, and 75th percentiles of the transaction amount distribution. The analysis focuses on how well the predicted probabilities align with

[2] Snyder, J. A., & Golladay, K. A. (2024). It Happened Again: Differences Between Single and Repeat/Poly-Victimization Among Financial Fraud Victims. Journal of White Collar and Corporate Crime, 5(1), 46-57. https://doi.org/10.1177/2631309X231195801

the observed probabilities across these transaction size categories.

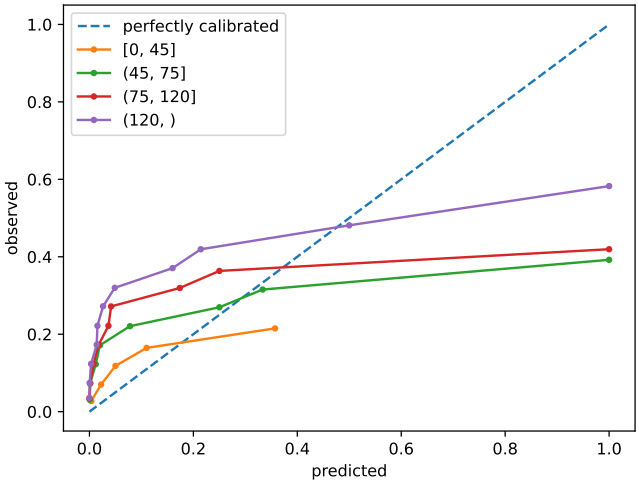


Figure 2: The reliability diagram by transaction amount groups (based on October data).

Figure 2 provides a clear illustration that the predicted fraud probabilities exhibit a tendency to overestimate probabilities near zero values and underestimate probabilities near one. Additionally, it is apparent that the predicted probabilities perform more effectively near zero values for smaller transactions and near one for larger transactions. This discrepancy suggests that the predicted probabilities are not uniformly effective across different groups of transactions. Consequently, the separate calibration efforts are necessary for each distinct group of transactions to ensure that the predicted probabilities align more accurately with the observed probabilities within each size category.

The calibration is done using Regression of observed on estimated values (ROE) method. This method allows for adjusting a given predicted value, Y_{pred}, to obtain a corrected value, Y_{ROE}:

$$Y_{ROE} = \alpha_{ROE} + \beta_{ROE} * Y_{pred}$$

The coefficients, α_{ROE} and β_{ROE}, are estimated by fitting linear regression with observed values, Y_{obs}, as the dependent variable and predicted values, Y_{pred}, as the independent variable. Essentially, the objective is to identify coefficients that can modify predicted values to more accurately represent observed values.

The calibration process is executed for each transaction group on a monthly basis in the following manner: the historical data on predicted fraud probabilities is grouped into probability bins with a width of 5%. Within each bin, both observed and predicted fraud probabilities are computed. Before fitting the linear regression, the dependent variable undergoes log-normalization. This adjustment is necessary because, as depicted in Figure 2, the relationship between predicted and observed values is not linear but rather logarithmic.

The coefficients derived from the linear regression are then utilized to calibrate the predicted fraud rates. Subsequently, the resulting calibrated values were transformed back into probabilities by applying the exponential function. To ensure the final calibrated probabilities remain within the valid probability range, the values were capped at 1.

Note that the calibration of fraud probabilities occurs subsequent to the adjustment of customer probabilities. The historical data on customer probabilities is adjusted as well to establish a suitable foundation for calibration. This sequential approach ensures that the calibration process is applied to a dataset that has been appropriately adjusted and reflects the interplay between customer and fraud probabilities.

4 Results

In this section we compare the performances of three models. The best one is further tested on the second batch data and used to draw conclusions and recommendations.

Model 1 is a baseline that contextualizes the results of trained models and acts as a reference. The baseline model uses an optimization model without updating probabilities based on historical data. The model 2 introduces customer probability adjustment to the optimization model, allowing it to learn from historical data and prioritize transactions based on customer behavioral patterns. The model 3

incorporates calibration in addition to the features of the model 2, refining the predictive accuracy over time by learning from past performances.

4.1 All information is available

Models comparison

The first batch data was used to evaluate three models. The results summarized in Tables 1, 2 and 3 (detailed results are presented in Appendix 6.1):

	Model 1	**Model 2**	**Model 3**
Success rate, cases	21.2%	32.6%	75.5%
Success rate, £	48.6%	62.9%	77.3%
External investigations, cases	500	500	108
External investigations cost, £	£50,000	£50,000	£10,800
Money saved, £	£123,001.8	£159,360.4	£195,748.4

Table 1: Overall performance of three models

	The number of investigated cases			**Success rate, cases**		
Priority	**Model 1**	**Model 2**	**Model 3**	**Model 1**	**Model 2**	**Model 3**
Priority 1	2,664	3,430	33,163	8.7%	15.5%	78.7%
Priority 2	4,274	4,783	3,982	38.8%	58.0%	65.0%
Priority 3	12,456	11,923	6,782	49.3%	71.6%	69.9%
Priority 4	246	289	344	48.2%	65.2%	80.1%
Total	19,640	20,425	44,271	21.2%	32.6%	75.5%

Table 2: The performance of three models by priority

	The number of investigated cases			**Success rate, cases**		
Bank	**Model 1**	**Model 2**	**Model 3**	**Model 1**	**Model 2**	**Model 3**
Bank A	3,813	3,364	7,722	19.0%	30.0%	71.9%
Bank B	4,746	5,483	11,487	22.1%	34.1%	77.2%
Bank C	3,525	3,940	8,494	18.1%	33.7%	75.8%
Bank D	3,245	3,305	7,265	20.9%	30.8%	73.5%
Bank E	4,321	4,333	9,303	24.2%	32.2%	75.8%
Total	19,640	20,425	44,271	21.2%	32.6%	75.5%

Table 3: The performance of three models by home bank

	The number of investigated cases			**Success rate, cases**		
Amount bins	**Model 1**	**Model 2**	**Model 3**	**Model 1**	**Model 2**	**Model 3**
[£0, £45]	0	7	23,328	0%	0.6%	76.6%
(£45, £75]	16,234	15,821	8,923	52.3%	73.0%	71.1%
(£76, £120]	1,576	2,209	9,172	24.1%	43.0%	83.3%
(£120,)	1,830	2,388	2,848	28.6%	48.0%	66.7%
Total	19,640	20,425	44,271	21.2%	32.6%	75.5%

Table 4: The performance of three models by transaction amount

Table 1 clearly indicates that Model 3 stands out as the superior choice, as evidenced by its highest success rates in identifying fraudulent cases and saving money. This underscores the importance of the learning component in achieving optimal performance. While adjusting customer probabilities proved effective in distinguishing between scam and non-scam cases, it did not contribute to minimizing the cost of investigation. On the other hand, Model 3, with calibrated fraud probabilities, not only enhanced performance but also reduced the cost of investigation.

While the number of external investigations has decreased by around 80% for Model 3, Table 2 further reveals that the number of investigated cases nearly doubled. By better assessing expected losses, the model has re-allocated resources from more time-consuming investigations to less time-intensive ones, without sacrificing success rates; in fact, success rates increased.

A noteworthy observation from Table 4 is that Models 1 and 2 fails to detect fraudulent activity in low-value transactions (amounting to £45 or less). However, with the introduction of calibration into the model, there is an improvement in performance. This enhancement can be attributed to the improved evaluation of expected losses, indicating that the calibrated model is better equipped to navigate and identify fraudulent activities in low-value transactions. This underscores the importance of fine-tuning and calibration processes in refining the model's sensitivity across different transaction categories, especially those involving lower monetary values.

Finally, comparing the models' performance by bank it is clear that all models work equally well for all banks. The number of investigated cases are proportional to the sizes of banks and the success rates does not deviate significantly from mean value.

Second batch data

The final adjustments of customer probabilities and calibration of fraud probabilities are performed at the end of the first batch data period.

	Model 3
Success rate, cases	72.9%
Success rate, £	55.0%
External investigations, cases	0
External investigations cost, £	£0
Money saved, £	£192,495.45

Table 5: The performance of Model 3 on second batch data

	The number of investigated cases	**Success rate, cases**	**Success target, cases**
Priority 1	15,837	82.1%	50%
Priority 2	1,672	58.3%	60%
Priority 3	2,496	53.9%	70%
Priority 4	178	77.9%	80%
Total	20,183	72.9%	

Table 6: The performance of Model 3 on second batch data by priority

Home Bank	**Number of investigated cases**	**Success rate, cases**
Bank A	3,627	63.4%
Bank B	5,250	78.1%
Bank C	3,871	74.1%
Bank D	3,312	73.7%
Bank E	4,123	72.1%
Total	20,183	72.9%

Table 7: The performance of Model 3 on the second batch data by home bank

Amount bin	**Number of investigated cases**	**Success rate, cases**
[£0, £45]	8,909	79.8%
(£45, £75]	3,291	58.3%
(£76, £120]	6,759	87.4%
(£120,)	1,224	64.6%
Total	20,183	72.9%

Table 8: The performance of Model 3 on the second batch data by transaction amount

Tables 5 and 6 reveal that the performance of the trained model on the test set is predictably inferior to the training set, evident in both the amount saved and the detection of fraudulent cases. An important observation is that the average amount of scam transactions is notably higher in the test set, amounting to £348, compared to the training set, which stands at £102. This disparity could provide an explanation for the significant decline in performance, particularly in terms of the money saved metric. The increased average amount for scam transactions in the test set likely poses a more challenging scenario for the model, influencing its ability to achieve comparable levels of savings as observed in the training set. Understanding and addressing such shifts in transaction characteristics are pivotal for enhancing the model's adaptability and maintaining effective performance across different datasets. The target success is achieved only for Priority 1 transactions, while the performance is sufficiently close to the target for Priority 2 and 4. However, for Priority 3, the model significantly underperforms and falls short of meeting the target. Despite these challenges, the model continues to effectively utilize information acquired during the training stage, and notably, no external investigators are employed.

Examining Table 7, it becomes apparent that the model's performance across different banks is no longer uniform. Notably, the success rate for Bank A is considerably lower compared to other banks. A potential explanation for this discrepancy could be the higher fraud rate observed for Bank A in the test set (0.6%) compared to the training set (0.5%), whereas other banks exhibit lower fraud rates on the test set. This underscores the importance of regular calibration for the model to ensure that historical data is consistently incorporated and effectively utilized. Regular calibration becomes crucial for adapting to evolving patterns and maintaining optimal performance across diverse banking environments.

Examining Table 8, it is evident that the model continues to accurately evaluate expected losses. This is particularly apparent in its ability to adequately cover low-value transactions. Despite potential challenges in other aspects of performance, the model's proficiency in assessing expected losses, especially for transactions with lower monetary values, is a positive aspect that highlights its robustness in managing different risk scenarios.

4.2 Only information about investigated cases is available

The next phase involves assessing the Model 3's performance in real-world scenarios where only information pertaining to investigated cases is accessible. This step is crucial since in practical situations, determining whether a transaction is a scam or not is uncertain until a thorough investigation is conducted.

Training set

	Model 3
Success rate, cases	40.7%
Success rate, £	45.5%
External investigations, cases	50
External investigations cost, £	£5,000
Money saved, £	£115,170.7

Table 9: The performance of Model 3 with limited information

	Number of investigated cases	**Success rate, cases**
Priority 1	9,522	39.9%
Priority 2	1,546	36.4%
Priority 3	3,100	43.3%
Priority 4	226	56.0%
Total	14,394	40.7%

Table 10: The performance of Model 3 by priority with limited information

Home Bank	**Number of investigated cases**	**Success rate, cases**
Bank A	2,388	39.9%
Bank B	3,957	43.2%
Bank C	2,620	37.6%
Bank D	2,338	41.9%
Bank E	3,091	39.4%
Total	14,394	40.7%

Table 11: The performance of Model 3 with limited information by home bank

Amount bin	**Number of investigated cases**	**Success rate, cases**
[£0, £45]	7,694	42.8%
(£45, £75]	4,039	44.3%
(£76, £120]	1,695	34.7%
(£120,)	966	35.5%
Total	14,394	40.7%

Table 12: The performance of Model 3 by transaction amount with limited information

In the limited information case, Model 3 exhibits an expected decrease in performance, with the success rate dropping from 75% to 40%. However, a notable positive aspect is that the model maintains stability. Success rates across different banks show minimal deviation, indicating that the model consistently learns not to allocate significant resources to time-consuming Priority 3 transactions. Consequently, it invests more effort in investigating Priority 1 transactions.

Despite the overall drop in success rate, the model's performance remains stable across different banks. Additionally, it continues to effectively cover transactions of all sizes. This suggests that even in scenarios with limited information, the model retains its ability to make informed decisions, allocating resources judiciously and adapting to different priorities and transaction characteristics.

Test set

	Model 3
Success rate, cases	39.9%
Success rate, £	43.0%
External investigations, cases	0
External investigations cost, £	£0
Money saved, £	£150,763.25

Table 13: Overall performance of Model 3 on test data with limited information

	The number of investigated cases	**Success rate, cases**	**Success target, cases**
Priority 1	20,061	41.0%	50%
Priority 2	524	27.2%	60%
Priority 3	1,141	37.9%	70%
Priority 4	98	55.9%	80%
Total	21,824	39.9%	

Table 14: The performance of Model 3 on test data by priority with limited information

Home Bank	**Number of investigated cases**	**Success rate, cases**
Bank A	4,573	33.5%
Bank B	5,142	44.1%
Bank C	4,244	38.5%
Bank D	3,639	42.7%
Bank E	4,226	38.5%
Total	21,824	39.9%

Table 15: The performance of Model 3 on test data by home bank with limited information

Amount bin	**Number of investigated cases**	**Success rate, cases**
[£0, £45]	18,585	41.1%
(£45, £75]	1,464	39.7%
(£76, £120]	1,439	41.1%
(£120,)	336	29.2%
Total	21,824	39.9%

Table 16: The performance of Model 3 on test data by transaction amount with limited information.

Again the last adjustments of customer probabilities and calibration of fraud probabilities are done at the beginning of test period using only information from the training set.

Upon testing the model on the test set, a consistent pattern is observed, with the performance for Bank A once again registering lower compared to all other banks. Despite this discrepancy, the model maintains overall stability across different priorities and transaction sizes.

It is noteworthy that the model exhibits a stable behavior, adhering to the priorities and sizes of transactions, even if the predefined success targets have not been fully achieved. This consistency in its approach across various scenarios suggests that the model, while not meeting all success criteria, remains robust and reliable in its decision-making process. This resilience could be crucial in real-world applications where external factors and data variations may impact performance.

5 Conclusion

In this report, we deployed a strategy consisting of a decision optimization model, a dynamic probability threshold for filtering inputs to the decision model, an adjustment mechanism to customer probabilities and a probability distribution calibration procedure. We compared three combinations of these approaches for learning from the past data. Both the customer probability adjustment and the overall probability calibration demonstrated positive impacts on the precision of detecting suspicious transactions.

However, probability calibration is the only approach that is able to identify detect frauds with small amounts based on our decision algorithm because it adjusts the expected loss for each transaction. We studied the performance of the established model using extra data and practical limitations on available information. While the performance dropped as expected, we were able to observe stability from the model.

6 Appendix

6.1 Detailed performance results of models 1, 2 and 3

- Model 1 - Baseline case.

	Fraud cases	Not fraud cases
Investigated	526	19 114
Not investigated	1,955	216 562
Success rate	21.2%	

	Fraud	Not Fraud
Investigated	£123,001.8	£6,330,766.26
Not investigated	£130,138.6	£26,389,499.12
Success rate	48.6%	

- Model 2 - Model with customer probability adjustment.

	Fraud cases	Not fraud cases
Investigated	810	19,615
Not investigated	1,671	216,061
Success rate	32.6%	

	Fraud	Not Fraud
Investigated	£159,360.4	£6,060,789.95
Not investigated	£93,780.0	£26,659,475.43
Success rate	62.9%	

- Model 3 - Model with customer probability adjustment and fraud probability calibration.

	Fraud cases	Not fraud cases
Investigated	1,872	42,399
Not investigated	609	193,277
Success rate	75.5%	

	Fraud	Not Fraud
Investigated	£195,748.35	£4,498,185.69
Not investigated	£57,392.05	£28,222,079.69
Success rate	77.3%	

- Model 3 - Model with customer probability adjustment and fraud probability calibration on out-of-time sample

	Fraud cases	Not fraud cases
Investigated	733	19,450
Not investigated	273	103,067
Success rate	72.9%	

	Fraud	Not Fraud
Investigated	£192,495.45	£1,814,463.01
Not investigated	£157,832.8	£15,190,418.94
Success rate	55%	

- Model 3 - Model with customer probability adjustment and fraud probability calibration given only information about investigated cases is available

	Fraud cases	Not fraud cases
Investigated	1,011	13,383
Not investigated	1,470	222,293
Success rate	40.7%	

	Fraud	Not Fraud
Investigated	£115,170.7	£1,689,131.2
Not investigated	£137,969.7	£31,031,134.18
Success rate	45.4%	

- Model 3 - Model with customer probability adjustment and fraud probability calibration on test set given only information about investigated cases is available

	Fraud cases	Not fraud cases
Investigated	401	21,423
Not investigated	605	101,094
Success rate	39.9%	

	Fraud	Not Fraud
Investigated	£150,763.25	£1,185,101.21
Not investigated	£199,565.0	£15,819,780.74
Success rate	45.4%	